

Critical Analysis: Boosting Language Models’ Reasoning with Chain-of-Knowledge Prompting

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1 Methodological Strengths

1.1 End-to-end framing of elicitation, verification, and rethinking

The paper presents an integrated framework that treats reasoning as a three-stage pipeline: Chain-of-Knowledge (CoK) prompting for structured rationale elicitation, F2-Verification for reliability estimation, and a rethinking process for iterative correction. This framing goes beyond modifying prompts and instead defines a complete loop from input to verified output. The methodology explicitly recognises hallucinated rationales as a first-class problem and designs mechanisms that act both before generation (prompt design) and after generation (verification and rethinking).

1.2 Structured Chain-of-Knowledge prompting

A central contribution is the CoK prompting format, which decomposes the rationale into:

- **Evidence triples (CoK-ET)**: a list of structured knowledge triples (s, r, o) capturing atomic facts relevant to the query and answer, such as (*Derrick White, isA, basketball player*).
- **Explanation hints (CoK-EH)**: free-form textual reasoning that narrates how the evidence supports the final answer.

This design has several methodological advantages.

- It enforces a separation between factual content (triples) and narrative reasoning (hints), which makes rationales more amenable to automatic checking than plain chain-of-thought text.
- It aligns the intermediate representation with standard knowledge graph structures, enabling reuse of existing tools from knowledge base completion and retrieval for verification.
- It encourages the language model to expose explicit task-relevant knowledge instead of relying solely on opaque internal representations, which improves interpretability.

The exemplars in Figure 1 and Table 6 demonstrate that the model learns to produce concise and task-specific triples for both commonsense and symbolic reasoning (for instance, Last Letter Concatenation).

1.3 Careful construction of CoK exemplars

The exemplar construction process is methodologically deliberate and multi-stage.

1. For each dataset, a set of labelled questions is sampled as demonstrations, following earlier work on in-context learning.
2. Zero-shot chain-of-thought prompting with the instruction “Let us think step by step.” is used to generate initial textual rationales, which serve as candidate CoK-EH.
3. A composite knowledge base is built from six domains, including dictionary, commonsense, entity, event, script, and causality knowledge. It aggregates Wiktionary, ConceptNet, Wikidata5M, ATOMIC, GLUCOSE, ASER, and CausalBank.
4. A sentence encoder and a maximum inner product search tool are used to retrieve candidate triples from this composite knowledge base given the textual rationale.
5. Five expert annotators refine the retrieved triples into final evidence triples for each exemplar, optionally assisted by a small self-training scheme that bootstraps from two manually annotated examples.

This pipeline grounds CoK-ET simultaneously in language model rationales and curated knowledge graph entries. The use of retrieval plus expert filtering is a reasonable compromise between fully manual annotation and fully automatic extraction and yields exemplars that tightly couple questions, triples, hints, and answers.

1.4 Dual-view F2-Verification for reliability scoring

The F2-Verification module introduces a principled way to score the reliability of a generated rationale using two views:

- **Factuality view:** each generated triple $(\hat{s}_{ij}, \hat{r}_{ij}, \hat{o}_{ij})$ is scored against the composite knowledge base. When the triple exists in the knowledge base, exact matching gives a factuality score. When it does not, a TransR-based knowledge graph completion model assigns a soft plausibility score using the standard translation in a relation-specific space.
- **Faithfulness view:** the explanation hint $\hat{H}_i$ is compared to the concatenation of the query, evidence triples, and answer, denoted $\hat{H}'_i = [\hat{Q}_i; \hat{T}_i; \hat{A}_i]$, using a SimCSE sentence encoder. The cosine similarity of their embeddings serves as a faithfulness score.

For each query, F2-Verification aggregates these signals into a scalar reliability score

$$C_i = \gamma \cdot \frac{1}{|\hat{T}_i|} \sum_j f_v(\hat{r}_{ij} \mid \hat{s}_{ij}, \hat{o}_{ij}, K) + (1 - \gamma) \cdot f_u(\hat{H}_i \mid \hat{H}'_i),$$

with a balancing coefficient $\gamma \in (0, 1)$ set to 0.5 in experiments. This score cleanly combines average factuality across triples with semantic faithfulness of the explanation.

The design is methodologically strong for three reasons. First, it clearly distinguishes factual correctness from textual alignment. Second, it produces a continuous reliability signal that can be used both for evaluation and as a control variable in downstream procedures. Third, it leverages off-the-shelf components (TransR and SimCSE) rather than requiring new large models to be trained.

1.5 Rethinking algorithm driven by reliability

The rethinking process uses the reliability score to decide whether a query should be revisited. All test queries start in an “unreliable” set. For each iteration:

1. The prompt combines the fixed CoK exemplars with the current query and elicits new evidence triples, explanation hints, and an answer.
2. F2-Verification computes $C_i^{(n)}$ for each query.
3. Queries with $C_i^{(n)} \geq \theta$ are considered reliable and removed from the unreliable set.
4. For the remaining queries, low-scoring triples are identified and replaced with correct triples from the knowledge base in the next prompt, providing targeted knowledge injection.

Figure 8 shows that most performance gains arise in the first three iterations and that appropriate choices of the threshold θ improve both commonsense (CSQA) and factual (BoolQ) accuracy. This provides empirical evidence that reliability-guided rethinking is an effective mechanism for self-correction.

1.6 Broad and competitive experimental evaluation

The experimental design is broad and uses strong baselines.

- **Task coverage.** The method is evaluated on twelve reasoning datasets: six commonsense and factual tasks (CSQA, StrategyQA, OpenBookQA, ARC-c, a sports understanding subset of BIG-Bench, BoolQ), two symbolic reasoning tasks (Last Letter Concatenation and Coin Flip), and four arithmetic reasoning tasks (GSM8K, SVAMP, AQuA, MultiArith).
- **Model backbones.** Experiments include text-davinci-002 and gpt-3.5-turbo, with additional CSQA and GSM8K results on gpt-4. This demonstrates that CoK is not tailored to a single model.
- **Baselines.** Comparisons cover standard few-shot and zero-shot prompting, zero-shot CoT, Manual CoT, Auto-CoT, complexity-based prompting (ComplexCoT), and combinations with Self-Consistency (SC).

On text-davinci-002, CoK consistently improves over Manual CoT and Auto-CoT across tasks. For instance, on CSQA accuracy increases from 73.5 (Manual CoT) and 74.4 (Auto-CoT) to 75.4 with CoK and 77.3 with CoK plus F2-Verification. On GSM8K, CoK improves accuracy from 46.9 (Manual CoT) and 47.9 (Auto-CoT) to 51.2. On MultiArith, CoK reaches 94.6 accuracy, higher than both Manual CoT (91.7) and Auto-CoT (92.0). On gpt-3.5-turbo, Table 1 and Figure 5 indicate that CoK and CoK with SC outperform Manual CoT and ComplexCoT on most tasks, and CoK with SC and F2-Verification is strongest for the tasks where a knowledge base and F2-Verification are available.

1.7 Systematic ablations and analyses

The paper includes several ablation and analysis studies.

- **Component ablations.** Removing evidence triples, explanation hints, factuality verification, or faithfulness verification leads to consistent drops in accuracy (Figure 3), confirming that all components contribute meaningfully.
- **Wrong exemplars.** Injecting different proportions β of incorrect triples into the demonstrations shows only mild performance degradation even when all triples are wrong (Figure 4). This reproduces and extends the observation that model accuracy can be weakly coupled to rationale correctness.

- **Domain adaptation.** Using GSM8K exemplars as prompts for commonsense tasks still yields reasonable performance (Table 2), showing that the CoK paradigm transfers across domains even when the content of exemplars is mismatched.
- **Prompt engineering.** Figure 6 compares manual and automatic construction of CoK-EH and CoK-ET on GSM8K. The combination of manual explanation hints with automatic evidence triples performs competitively, suggesting that some annotation effort can be saved by automatic retrieval.
- **Faithfulness scoring.** Figure 7 compares SimCSE, FactCC, and Knowledge F1 as faithfulness scores. SimCSE yields the highest accuracy on most commonsense tasks, indicating that it is a reasonable choice for hallucination reduction.
- **Knowledge injection timing.** Table 5 contrasts naive knowledge injection before reasoning with knowledge injection after F2-Verification. The examples show that post-verification injection leads to more focused use of knowledge and avoids distracting or irrelevant triples.

Taken together, these analyses provide a reasonably complete picture of how and why CoK improves reasoning performance.

2 Key Limitations

2.1 Heavy dependence on curated knowledge bases and annotation

The framework relies strongly on a composite external knowledge base and on manually constructed CoK exemplars.

- The evidence triples in demonstrations are produced by five expert annotators based on retrieved candidates. The paper does not report annotation time, agreement statistics, or cost, which makes the resource requirement opaque.
- The factuality component assumes that the composite knowledge base is sufficiently complete and accurate. In practice, many domains lack high-coverage knowledge graphs, and existing graphs can be outdated or noisy.
- F2-Verification is only applied to tasks where such a knowledge base is available. For arithmetic reasoning and the coin dataset, the framework falls back to CoK prompting without verification, which limits the generality of the full pipeline.

These dependencies restrict the direct applicability of CoK prompting to well-resourced domains and reduce reproducibility for researchers without access to the same knowledge bases.

2.2 Limited evaluation scope beyond short-form text reasoning

The evaluation focuses on short-form question answering benchmarks with single-turn queries.

- The tasks are mostly multiple-choice or short answer questions. Long-context reading comprehension, multi-document reasoning, and interactive dialogue scenarios are not evaluated.
- All datasets are in English. No experiments are conducted on multilingual or cross-lingual reasoning tasks, although the knowledge base and SimCSE are also predominantly English-centric.

- Multi-modal settings, such as vision-language reasoning, are not considered, although structured evidence formats such as CoK-ET could be particularly beneficial in those contexts.

As a result, the empirical conclusions are strongest for the chosen benchmarks and cannot be assumed to hold unchanged in more complex or heterogeneous settings.

2.3 Insufficient analysis of explanation quality and hallucination

Although the paper explicitly targets hallucinated and unfaithful rationales, the evaluation of explanation quality remains indirect.

- There is no large-scale human study comparing CoK explanations to standard CoT explanations on dimensions such as factual correctness, logical coherence, or usefulness for end users.
- Hallucination is analysed mainly through accuracy changes under different faithfulness scores and through the behaviour of F2-Verification. The work does not provide explicit hallucination rates based on human-labelled rationales.
- Failure modes where F2-Verification assigns high scores to flawed rationales, or low scores to correct but unsupported rationales, are not systematically examined.

This limits the strength of claims about improved trustworthiness of explanations, beyond improved benchmark accuracy.

2.4 Absence of computational and cost analysis

The framework adds non-trivial computational overhead.

- Knowledge retrieval from multiple large knowledge bases is required for each query during verification and during exemplar construction.
- The TransR and SimCSE models introduce additional forward passes per query.
- The rethinking process can involve multiple additional generations per query, and Self-Consistency further multiplies the number of sampled reasoning paths.

Despite these overheads, the paper does not report inference time, token usage, or monetary cost, nor does it compare cost-effectiveness against simpler alternatives such as using a stronger backbone model or a larger number of CoT samples. This omission makes it difficult to assess practical viability in deployment settings.

2.5 Residual coupling between CoK format and actual knowledge use

The negative replacement experiment in Figure 4 reveals that model accuracy can remain high even when all evidence triples in demonstrations are randomly replaced. This confirms earlier observations that language models can ignore flawed rationales and still predict correct answers.

Although CoK is designed to surface explicit knowledge, these results indicate that the model sometimes treats the CoK structure as a formatting pattern rather than deeply relying on the factual content of the triples. The paper acknowledges this phenomenon but does not propose concrete mechanisms to enforce stronger reliance on correct evidence.

3 Technical Bottlenecks

3.1 Triple-based representation and coverage limitations

Representing intermediate reasoning exclusively as (s, r, o) triples has inherent limitations.

- Many reasoning steps involve higher-arity relations, temporal constraints, or nested conditionals that are difficult to encode as flat triples without loss of information.
- Numerical and symbolic reasoning, which are central in arithmetic tasks, are not naturally expressed as knowledge graph triples and therefore receive limited benefit from CoK-ET.
- The dependence on existing knowledge graphs means that emerging or domain-specific concepts that are missing from the knowledge base cannot be properly verified.

These constraints create a technical bottleneck when applying CoK to domains where reasoning is not well captured by simple factual relations.

3.2 Limitations of embedding-based faithfulness verification

The faithfulness component uses SimCSE similarity between the explanation and the concatenated question, evidence, and answer. This approach has several technical weaknesses.

- SimCSE is trained for general sentence similarity, not for recognising whether a rationale faithfully reflects a specific step-by-step reasoning process.
- Semantic similarity can be high even when the explanation contains subtle logical errors, incorrect numerical values, or incorrect causal relations.
- The faithfulness model is independent of the underlying language model, which can lead to miscalibration: an explanation that is faithful to the model’s internal reasoning may still receive a low similarity score if phrased differently.

These factors introduce noise into the reliability score and cap the precision of F2-Verification.

3.3 Non-differentiable and decoupled optimisation

The entire framework operates in an in-context and post-hoc manner. The base language model parameters are frozen, and neither CoK prompting nor F2-Verification is used as a direct training signal.

This yields two key bottlenecks.

- The model is not optimised to maximise reliability scores. Any improvement in C_i arises indirectly from prompt design and from the inductive biases of the pretrained model, which limits the extent to which reliability can be improved.
- The verification module does not influence the internal representations of the language model. It only processes generated outputs, which constrains its ability to steer the underlying reasoning mechanisms.

As a result, the approach is more of a wrapper around existing models than a method for fundamentally improving their reasoning capabilities.

3.4 Scalability and convergence of the rethinking process

The rethinking algorithm raises several technical challenges.

- The number of generations per query grows with the iteration count and with the use of SC, which may be prohibitively expensive at large scale.
- The paper observes empirically that two or three iterations are usually sufficient, but does not provide theoretical guarantees of convergence or safeguards against oscillations in edge cases.
- The choice of reliability threshold θ strongly affects behaviour. Very low thresholds can trigger over-injection of knowledge, slightly degrading performance on some tasks due to the inclusion of distracting triples.

These issues mean that careful hyperparameter tuning is required and that the procedure may not be robust under distribution shift.

4 Research Implications

4.1 Prompting with structured evidence as a general design principle

The success of CoK relative to CoT indicates that prompting large language models with structured evidence plus textual hints is a promising paradigm. Several broader implications follow.

- Prompt formats that expose explicit evidence can serve as interfaces between neural language models and symbolic knowledge systems.
- Intermediate representations that separate facts from reasoning text enable post-hoc verification, editing, and knowledge injection, which are difficult when rationales are unstructured.
- Evidence-aware prompting may improve interpretability of model decisions, since each step of reasoning is backed by explicit triples that can be inspected.

4.2 Verification as a core component of reasoning pipelines

By demonstrating measurable gains from F2-Verification and rethinking, the paper strengthens the view that verification should be integrated into reasoning systems rather than treated as an optional add-on.

- Reliability scores derived from factuality and faithfulness provide a concrete signal for deciding when to trust a prediction, when to regenerate, and when to inject additional knowledge.
- The framework illustrates how external tools such as knowledge graph completion models and sentence encoders can be composed with language models without joint training.
- The hallucination analysis highlights that choosing an appropriate faithfulness metric has a direct impact on accuracy and robustness, implying that metric design is a key part of building trustworthy reasoning systems.

4.3 Revealed gap between rationale quality and prediction accuracy

The experiments on wrong exemplars reveal that even completely corrupted evidence triples in demonstrations do not cause catastrophic accuracy drops. This supports the broader claim that current benchmarks and metrics are only weakly sensitive to rationales and that models can succeed by exploiting superficial patterns in prompts.

This gap has two important implications.

- Benchmark accuracy alone is not a reliable indicator of correct reasoning. Evaluation protocols that ignore rationales can overestimate model reliability.
- Methods that improve rationale quality without dramatically changing accuracy, such as CoK with F2-Verification, are still valuable because they address trust and interpretability dimensions that are not captured by standard metrics.

4.4 Interaction with retrieval-augmented and tool-augmented models

The framework can be seen as a concrete instance of tool-augmented reasoning, where the language model interacts with an external knowledge base and specialised verification tools. This has implications for the design of future systems.

- Retrieval modules can be used not only to enrich prompts but also to validate and repair the model’s own generated evidence.
- Knowledge injection strategies that act after verification, rather than before reasoning, appear more precise and less distracting, as shown in the knowledge injection case study.
- Similar architectures could integrate additional tools for numeric computation, program execution, or formal reasoning, with CoK-style representations serving as the glue.

5 Potential Research Directions

5.1 Scalable and automated CoK exemplar construction

Reducing manual effort in constructing CoK exemplars is a natural next step.

- Large language models can be used to automatically propose candidate evidence triples given questions and CoT-style rationales, with humans only validating and editing a subset.
- Distant supervision techniques could align text corpora with knowledge graphs to infer triples at scale, which can then be filtered using simple heuristics or lightweight classifiers.
- Active learning can identify which exemplars are most valuable for improving performance, focusing annotation resources where they have the largest impact.

5.2 Learning-based verification and joint optimisation

F2-Verification currently uses fixed scoring functions. Future work could introduce parametric verification models and joint training.

- Neural verifiers trained on labelled pairs of rationales and correctness labels could replace or augment TransR and SimCSE, potentially capturing richer dependencies between evidence and explanations.

- Reliability scores could be integrated into fine-tuning objectives, encouraging the language model to generate rationales that are both accurate and easy to verify.
- Differentiable approximations of the verification process could provide gradients that shape the model’s internal reasoning behaviour rather than only its outputs.

5.3 Richer intermediate representations

Expanding beyond triple-based CoK-ET would address the expressiveness bottleneck.

- Graph-structured representations with typed edges, temporal annotations, and causal labels could better model multi-hop and temporal reasoning.
- Program-like intermediate languages, where each step corresponds to an executable operation, could connect CoK prompting with program synthesis and neural-symbolic execution.
- Hybrid representations combining natural language clauses with structured tags for entities, relations, and evidential status could balance expressiveness and verifiability.

5.4 Robustness to imperfect and dynamic knowledge bases

Further research is required to understand how CoK performs when knowledge bases are noisy, incomplete, or evolving.

- Controlled experiments can inject synthetic noise or remove subsets of triples to quantify robustness and identify fragile regions.
- Probabilistic reasoning over multiple candidate triples, rather than relying on single scores, could mitigate the impact of incorrect entries.
- Mechanisms for online knowledge base update, informed by interactions with the language model, could maintain alignment between the model’s beliefs and the verification source.

5.5 Cost-aware and adaptive rethinking strategies

The rethinking process could be redesigned to account explicitly for computational cost.

- Lightweight uncertainty estimates from the base model could decide when F2-Verification and rethinking are necessary, avoiding expensive processing for confidently correct cases.
- Multi-stage verification pipelines could deploy cheap checks first and invoke more expensive methods only when simpler checks are inconclusive.
- The selection of triples for correction and knowledge injection could be made more selective, focusing on evidence that strongly influences the final answer.

6 Conclusion

The Chain-of-Knowledge prompting framework provides a coherent methodology for improving large language model reasoning by combining structured evidence extraction, dual-view verification, and reliability-guided rethinking. The experimental results show consistent, although moderate, gains over strong chain-of-thought baselines across a wide range of reasoning tasks and model backbones. The work also highlights important phenomena, such as the loose coupling between rationale correctness and accuracy, and the benefits of targeted knowledge injection.

At the same time, the approach has clear limitations: it depends on extensive manual annotation and composite knowledge bases, it is evaluated primarily on short English reasoning benchmarks, it uses verification modules that are only approximate proxies for explanation correctness, and it introduces significant unquantified computational overhead. The framework operates as a sophisticated wrapper around fixed language models rather than as a training procedure that fundamentally reshapes their reasoning processes.

Overall, the paper offers a valuable step toward evidence-aware reasoning in large language models and opens multiple research directions in automated exemplar construction, learning-based verification, richer intermediate representations, robust knowledge integration, and cost-aware self-correction.